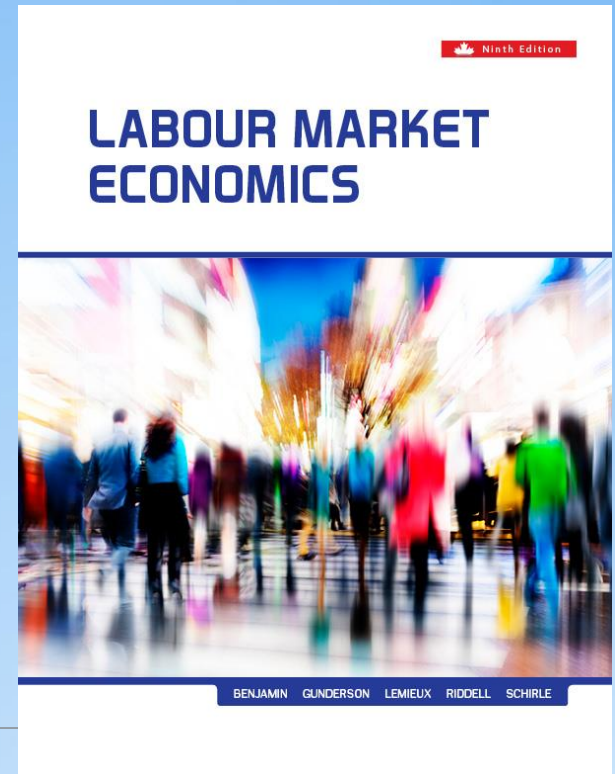


Chapter One

Introduction to Labour Market Economics



Prepared by Dr. Mir Ahasan Kabir
York University

Learning Objectives

1. Understand that earnings, hours, and hourly wages are quite unevenly distributed in the Canadian labour market.
2. See that, as in other markets, supply and demand curves can be used to find the equilibrium values of wages and employment levels.
3. Understand that, in addition to market forces, legislative and institutional factors play an important role in labour market outcomes.
4. Learn about the different types of data sets, such as time-series data, cross-section microdata, and panel data, that are used in labour market research.
5. Learn how regression analysis, and other econometric techniques, contribute to our understanding of labour economics.

Main Questions

- Who are the main actors in the labour market, and what particular roles do they play?
- How can features of the labour market be divided into labour supply and labour demand factors, as well as into the interaction of supply and demand?
- What types of policy questions does labour economics address?
- What characteristics of the labour market make it distinctive from other markets, justifying a special sub-discipline of economics?
- What is the neoclassical model of the labour market? Are there alternative approaches?

Main Players and their Roles

Individuals' Decisions

- When to enter the labour force
- How much education/training
- Type/extent of job search
- Occupation/industry
- Labour hours/household work
- Accept/quit/relocate
- Wage rate
- Union/association

Main Players and their Roles

Firms' Decisions

- Numbers of workers
- Wages/benefits
- Hours
- Layoff/bankruptcy
- Subcontract
- Pension/retirement policy

Main Players and their Roles

Factors Influencing the Firm's Decision

- **Global Competition:**

Free trade, Industrial restructuring, Deregulation, Privatization

- **Legislative Environment:**

Human Rights, Minimum wage, Overtime, Maternity leave, Worker's compensation, Occupational health/safety, Pensions/mandatory retirement, Labour laws/collective bargaining

- **Changing Workforce:**

Age, Gender, Ethnic diversity

Main Players and their Roles

Government's Decisions

- Individual rights vs. employer competitiveness
- Public support
- Training
- Insurance
- Compensation
- Pensions
- Vocational rehabilitation

Labour Market Economics

Involves analyzing the determinants of the various dimensions of *labour supply* and *demand* and their interaction in alternative market structures to determine wages, employment, and unemployment.

Labour Supply and Demand

Dimensions of Labour Supply

Quantity Dimensions:

Population Growth, Labour Force Participation...

Quality Dimensions:

Education, Health, Labour Productivity...

The Impact of Income Maintenance Programs:

Welfare, Disability, Employment Insurance...

Labour Supply and Demand

Dimensions of Labour Demand

Labour Costs and the Firm's Hiring Decisions:

The impact of quasi-fixed costs and legislatively imposed costs...

Demand for the Firm's Output and the Firm's Competitiveness:

Globalization, Industrial restructuring, Free trade, Technological advancements...

Labour Supply and Demand

Labour Market Outcomes:

Wages, Employment, Unemployment, Labour Shortages, and Wage Differentials...

Factors Influencing the Outcomes:

Legislative Interventions, Unions and Collective Bargaining, the Market Structures (in both the product and the labour markets)...

Preliminary Explorations of Labour Market Outcomes

Table 1.1: Sources of Income for Individual Canadians, 2017

	Full Sample		Men		Women	
	Mean	% positive	Mean	% positive	Mean	% positive
Earnings	41,348	82.6	50,824	86.2	31,910	79.0
Investment income	2,542	27.1	2,906	27.7	2,179	26.5
Pension income	1,262	4.0	1,484	4.3	1,040	3.7
Other income	952	14.2	971	13.7	933	14.7
Government transfers	3,728	60.1	2,495	52.8	4,955	68.5
Subtotal	49,831	98.1	58,681	98.6	41,017	97.6
Income taxes	8,391	70.0	11,168	75.2	5,625	64.8
After-tax total	41,440	98.2	47,512	98.7	35,392	97.8
Annual hours worked	1,411	81.1	1,593	85.3	1,230	76.8
Average hourly wage	29.77	73.3	32.57	76.7	26.67	69.8
Worked, but not self-employed	64.7		66.1		63.3	

Preliminary Explorations of Labour Market Outcomes

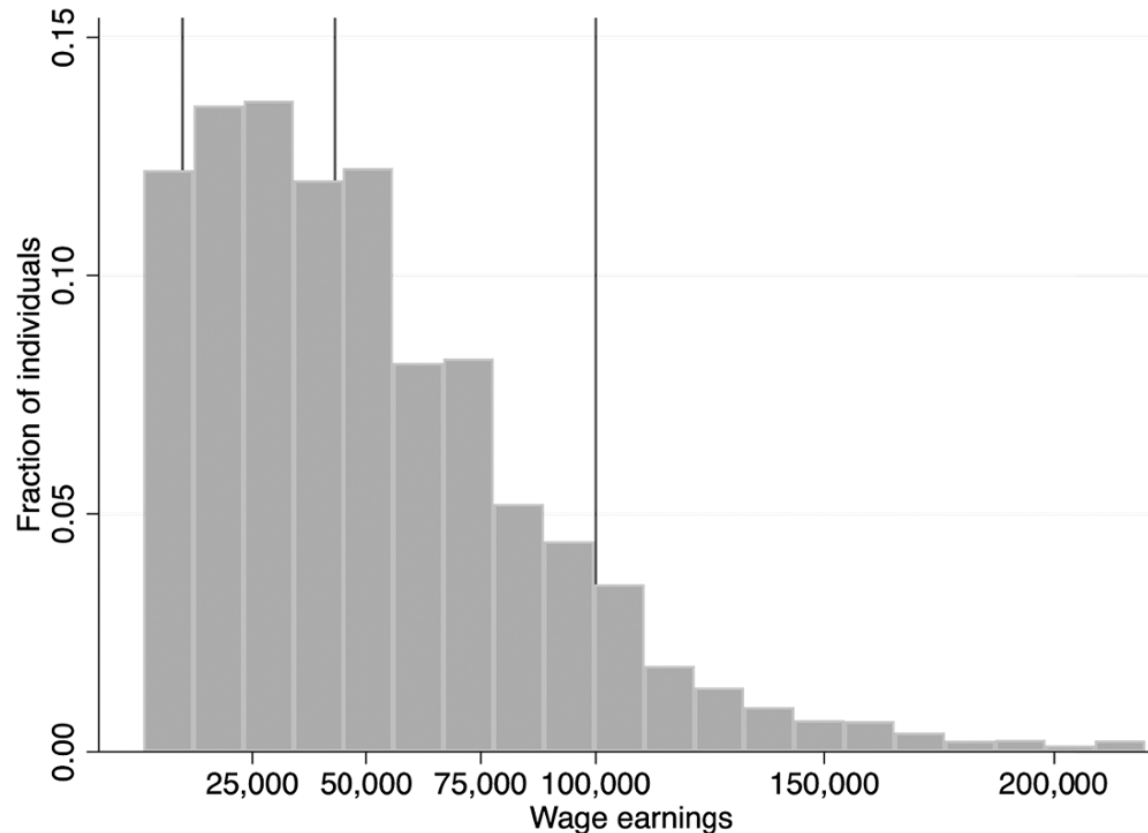


Figure 1.1 The Distribution of Individual Labour Earnings, 2017

Preliminary Explorations of Labour Market Outcomes

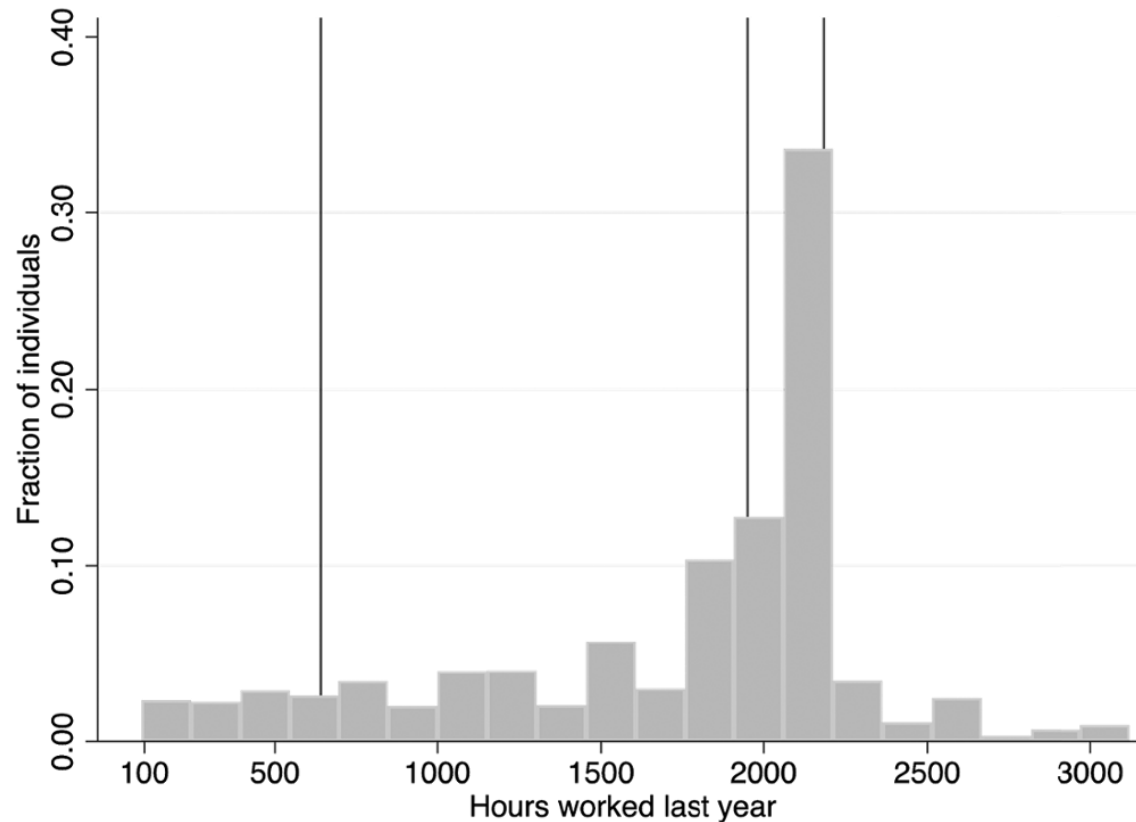


Figure 1.2 The Distribution of Individual Annual Hours Worked, 2017

Preliminary Explorations of Labour Market Outcomes

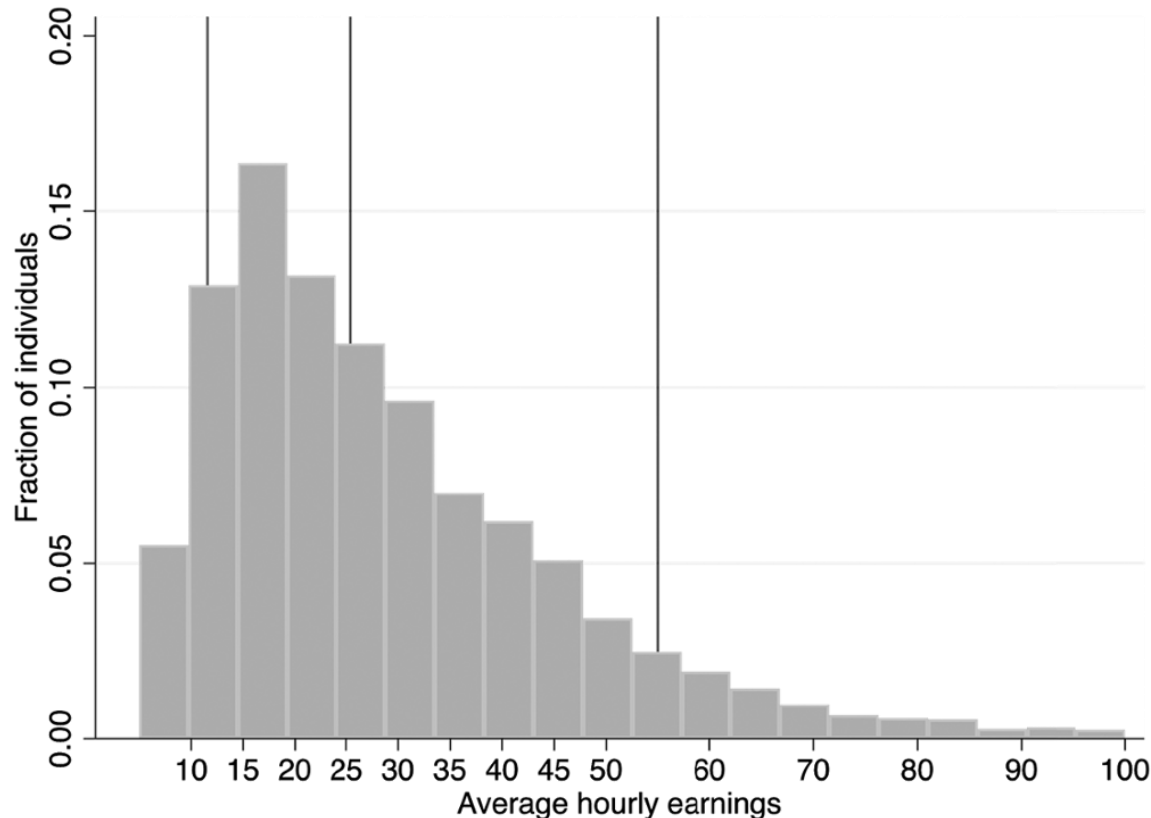


Figure 1.3 The Distribution of Individual Average Hourly Earnings (Wages), 2017

The Neoclassical Model of Labour Supply and Demand

Key Ingredients:

1. Behavioural assumptions about how buyers and sellers respond to prices and other factors.
2. Assumptions about how buyers and sellers interact, and how the market determines the level and terms of exchange.

Wages and Employment in a Competitive Labour Market

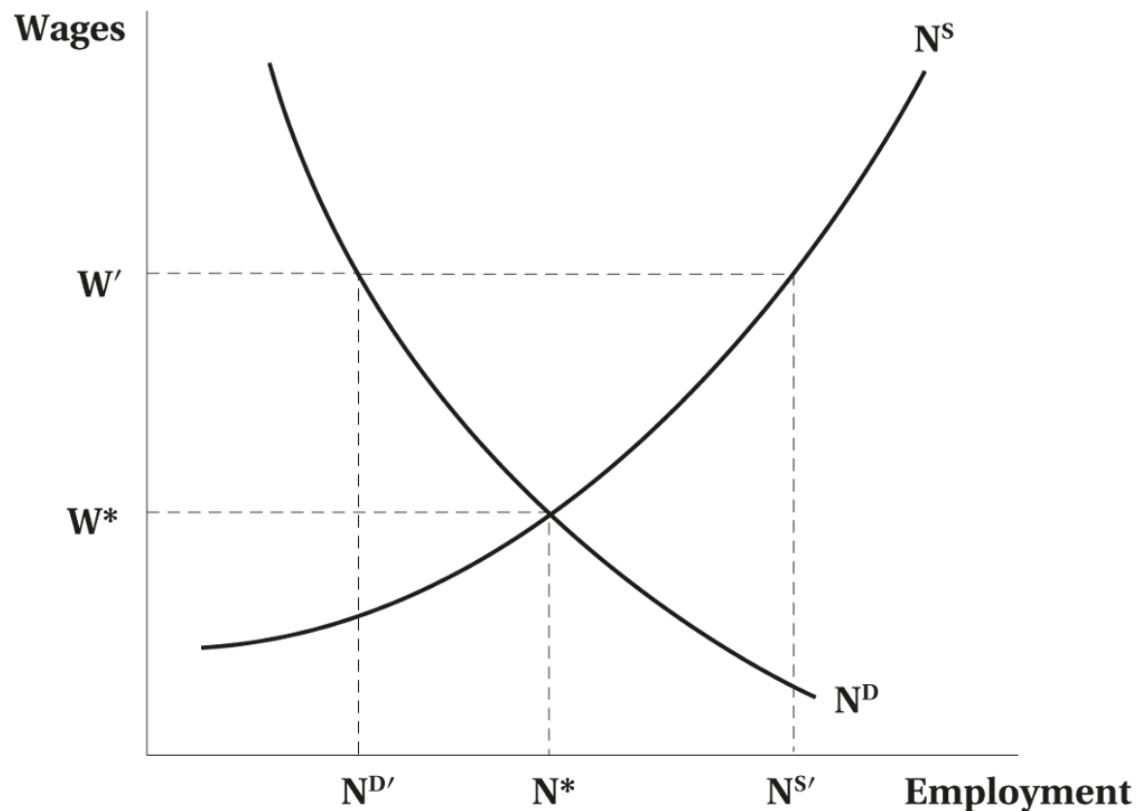


Figure 1.4 Wages and Employment in a Competitive Labour Market

Wages and Employment in a Competitive Labour Market

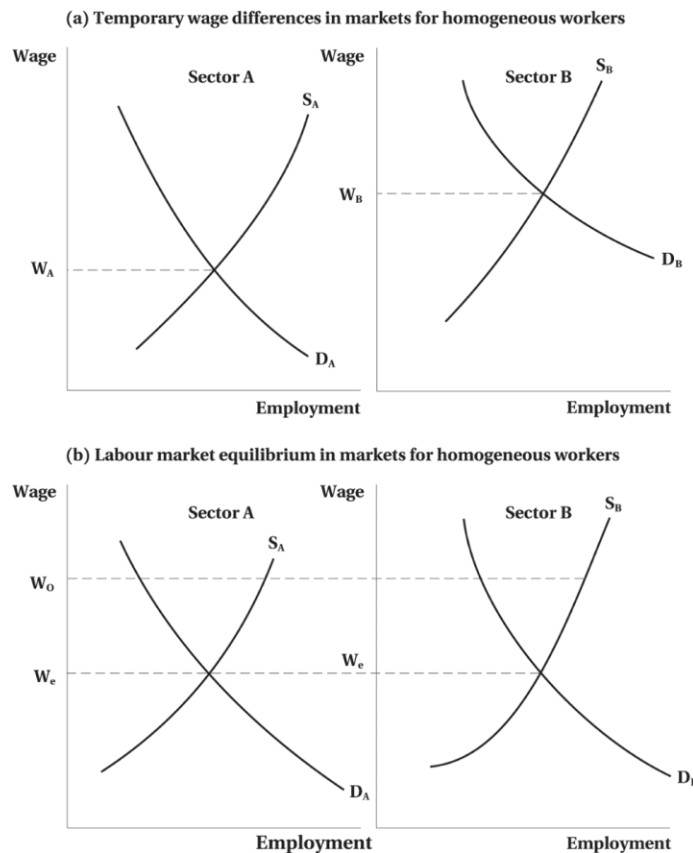


Figure 1.5 Equilibrium Wages across Two Sectors

Current Policy Issues

Labour market economics should be able to shed light on policy issues affecting:

- Labour Supply
- Labour Demand
- Wage Determination and Wage Structures
- Unemployment

Current Policy Issues

Labour Supply:

Work incentive effects of income maintenance and tax transfer programs such as employment insurance, welfare, subsidized day care...

Labour Demand:

The labour demand effects of globalization and outsourcing, trade policies, collective bargaining, legislative policies with respect to minimum wages, human rights, workplace health and safety...

Current Policy Issues

Wage Determination and Wage Structures:

Wage polarization, non-wage benefits, male-female wage differentials, pay equity legislation, retirement, and pensions...

Unemployment:

Changes in unemployment rate, job stability, impact of employment insurance on unemployment, youth unemployment...

COVID-19 and the Labour Market

COVID-19 would have a profound impact on the labour market, at least in the short run.

- Many workers were laid off and could not work from home.
- Others had their hours increase to deal with changes emanating from the pandemic.
- Many parents without childcare were unable to continue working their usual hours and were laid off as a result.

COVID-19 and the Labour Market

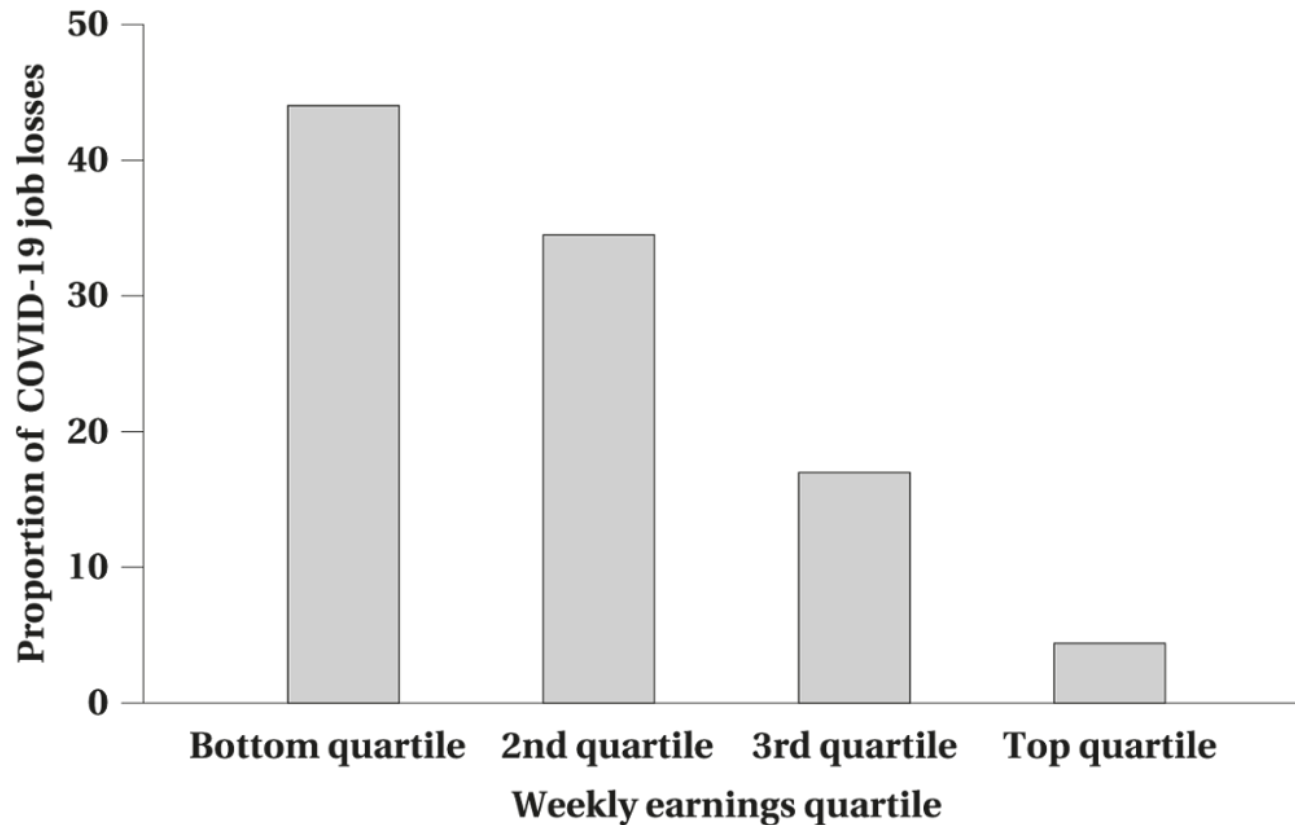


Figure 1.6 The Distribution of COVID-19 Employment Losses Across Earning Quartiles

Similarities and Differences between the Labour Market and Other Markets

- Various Actors with a Variety of Goals
- Sociological, Institutional, and Legislative Constraints
- Market Imperfections
- Complex Price, Serving a Variety of Functions

Chapter Summary

There are three main groups of actors in the labour market: individuals, firms, and government.

While there are many dimensions to the decisions made by these actors, we can distill the outcomes of the decisions to individual outcomes, especially concerning earnings, employment (or hours), and wages.

The neoclassical supply and demand model is a powerful tool that can be used to analyze these outcomes, especially employment and wages.

Most of this book is organized around the supply and demand framework.

REGRESSION ANALYSIS AND RESEARCH DESIGNS

Appendix 1A

Types of Data

Aggregate or Time-series Data

Cross-section Microdata

Panel or Longitudinal data

Supply and Demand and the Relative Earnings of “Star” versus “Grinder” Hockey Players

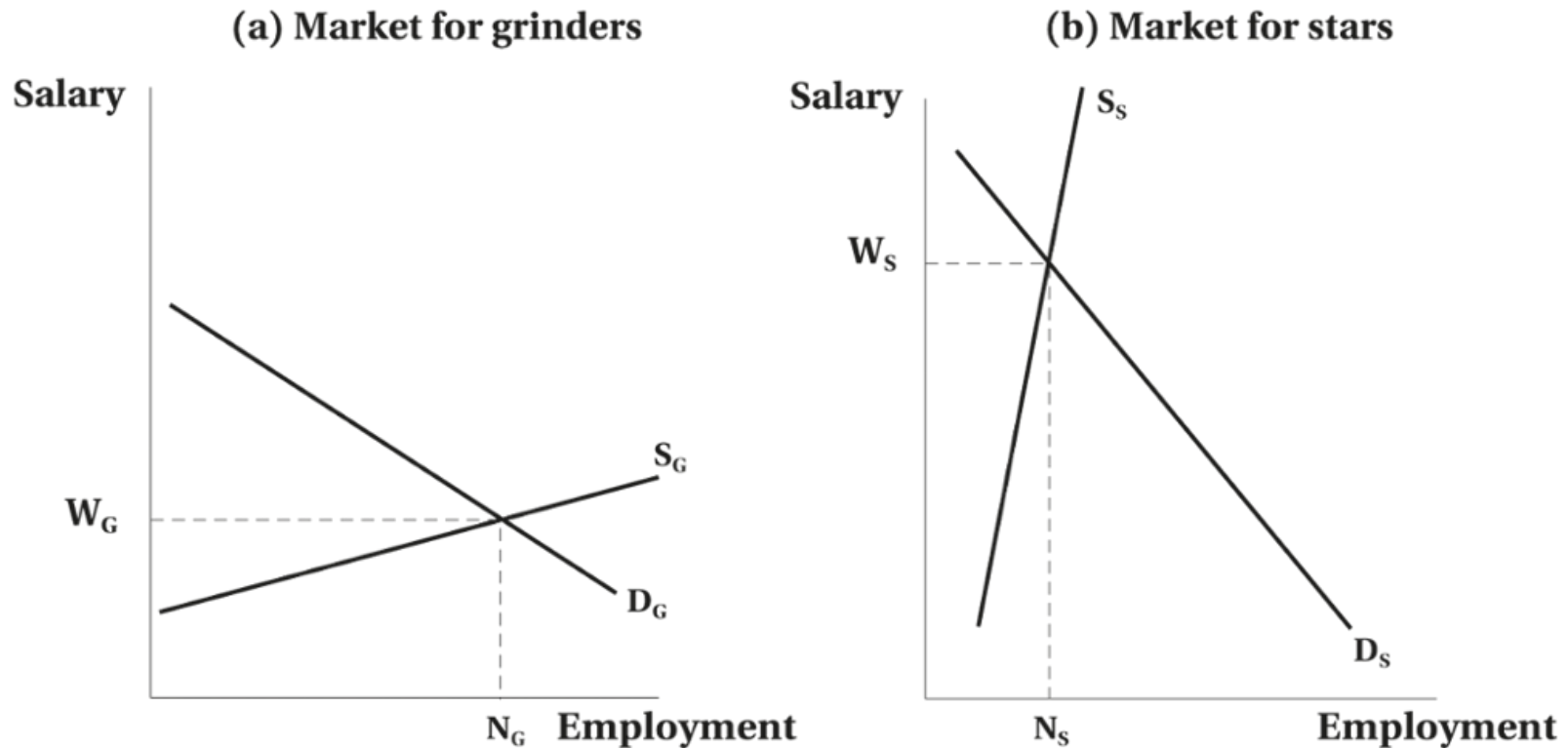


Figure 1.A1 Supply and Demand and the Relative Earnings of Star versus Grinder Hockey Players

NHL Player Salaries by Goals Scored in the Previous Year, 2018–19

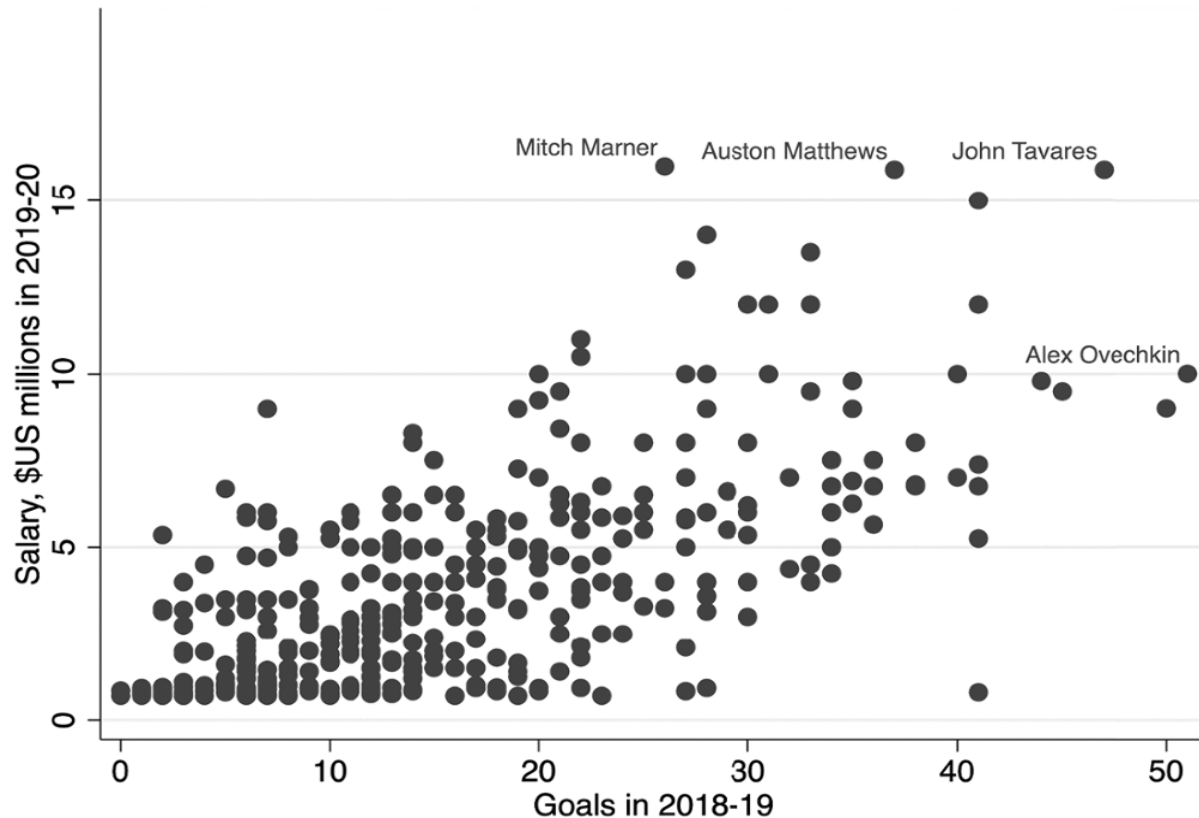


Figure 1.A2 NHL Player Salaries by Goals Scored in the Previous Year

Logarithm of NHL Player Salaries by Goals Scored in the Previous Year, and Fitted Regression Line, 2018–19

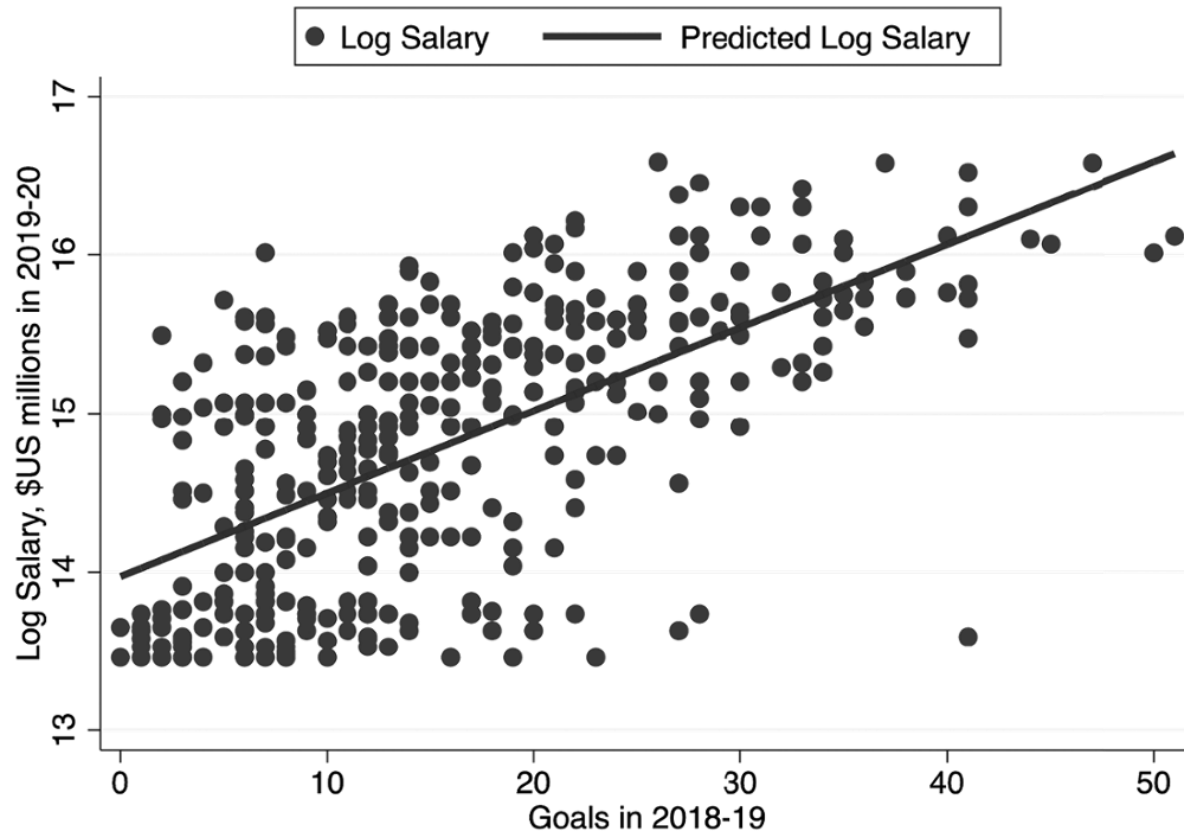


Figure 1.A3 Logarithm of NHL Player Salaries by Goals Scored in the Previous Year, and Fitted Regression Line

Estimating Effects of Performance on Player Salary in 2015–16

We can summarize the relationship between player's salary and goals as follows:

$$\text{Salary} = a + b \times \text{Goals}$$

$$\text{Salary} = a + b \times \text{Goals} + e$$

Ordinary Least Square (OLS) estimator

Tabulation of NHL Player Salaries by Goals Scored, 2018–19

TABLE 1.A 1: Tabulation of NHL Player Salaries by Goals Scored, 2018–2019

Goals Scored, 2018–2019	Sample Size	Salary (\$US millions)	In (Salary)
0–9	121	1.93	14.20
10–19	133	3.13	14.73
20–29	76	5.53	15.32
30–39	32	7.51	15.76
40–49	11	9.03	15.82
50 +	2	9.50	16.07

Estimated Effects of Performance on Player Salary in 2019–20 (OLS)

TABLE 1.A2 Estimated Effects of Performance on Player Salary in 2019–2020
(standard errors in parentheses)

		Salary (\$US) mean = 3,807,829			ln (Salary) mean = 14.81		
	Means	(1)	(2)	(3)	(4)	(5)	(6)
Intercept		685,592	247,395	197,563	13.972	13.867	13.859
		(218,701)	(202,844)	(211,570)	(0.064)	(0.062)	(0.064)
Goals	15.97	195,498	88,605	90,550	0.052	0.027	0.027
		(11,468)	(15,469)	(15,651)	(0.003)	(0.005)	(0.005)
Assists	21.84		98,218	98,902		0.024	0.024
			(10,569)	(10,605)		(0.003)	(0.003)
Plus/minus	−0.47			−8,156			−0.001
				(9,796)			(0.003)
R-squared		0.44	0.54	0.54	0.39	0.47	0.47

RESEARCH DESIGNS IN LABOUR ECONOMICS

Appendix 1B

Research Designs

Empirical Strategies for Estimating the Counterfactual Outcomes

The Distinction between Experimental and Non-experimental Methods

Experimental Method

Involve random assignment to “treatment” and “control” groups.

Have two advantages

- The statistical techniques needed are very simple and do not require additional assumptions.
- These methods are easy to explain to, and be understood by, policymakers and the general public.

Non-Experimental Method

Multiple regression is also often needed to control for potentially confounding factors.

1. Before–after comparison of those treated
2. Post-treatment comparison of treated and comparison groups
3. Difference-in-differences: before–after comparison with a comparison group
4. Natural experiments (or quasi-experiments)
5. Regression discontinuity designs